# Fast Surrogate Modeling of Excitable and Oscillatory FitzHugh-Nagumo Dynamics with Parametric Neural Operators

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**Abstract.** The FitzHugh-Nagumo (FHN) system serves as a simplified model of neuronal voltage dynamics, capturing the activator-inhibitor structure behind both isolated action potentials and the rhythmic spiking seen across the brain. Exploring its 5D physiological parameter space is important for neuromodulation and mapping voltage recordings back to biophysics, yet classical finite-difference solvers make rapid parameter sweeps expensive. We train parameter-conditioned Fourier Neural Operators (FNOs) as fast, differentiable surrogates for the FHN voltage and recovery fields on a one-dimensional spatial domain, conditioning each Fourier layer on the parameter vector $\boldsymbol{\lambda} = (D_u, D_v, a, b, \tau)$ via feature-wise linear modulation (FiLM). We apply a single bifurcation analysis that delimits the two distinct regimes the model spans, oscillatory (tonic firing) and excitable (action-potential propagation), and we train one operator in each. In the oscillatory regime the surrogate attains sub-$0.1\%$ relative $L^2$ error on both fields, runs nearly three orders of magnitude faster than the finite-difference baseline, generalizes uniformly across the parameter space, and extrapolates to low single-digit percentage errors outside of the training bounds. In the excitable regime the same operator accurately reproduces the firing threshold and the $c \propto \sqrt{D_u}$ conduction-velocity law and replicates full traveling pulses, fully capturing the excitable bifurcation structure rather than just smoothly interpolating fields.

**Keywords:** FitzHugh-Nagumo model · Excitable media · Fourier Neural Operators · Operator learning · Parametric PDEs · FiLM conditioning · Bifurcation analysis · Neuromodulation

## 1 Introduction

Much of the brain's signaling is shaped by two coupled processes inside each neuron: a fast change in membrane voltage that produces an action potential, and a slower recovery current that resets the cell [7,2]. The FitzHugh-Nagumo (FHN) reaction-diffusion system [5,18] reduces these dynamics into two coupled equations for membrane voltage and recovery, and remains a widely used reduced model of neural excitation by capturing both single action-potential propagation and the rhythmic spiking that dominates many human-brain signals. Connecting noisy voltage recordings back to the parameters of such a model is important for neural decoding, inverse modeling, and closed-loop neuromodulation.

These are all tasks that demand many fast, differentiable forward solves, which traditional finite-difference or spectral methods [2,1] cannot provide because they require fine discretization and become prohibitive over large parameter sweeps. For parametric PDEs, the cost of classical solvers grows quickly with the parameter space, since each new parameter combination requires a full re-simulation. This makes tasks such as inverse parameter inference (fitting $\boldsymbol{\lambda}$ to a voltage recording) and stimulus optimization for closed-loop control challenging, and motivates fast learned surrogates. Neural Operators (NOs) [11] learn mappings between infinite-dimensional function spaces, capturing the solution operator itself rather than a discretization, and Fourier Neural Operators (FNOs) [13] are particularly effective for PDEs by performing their convolutions in the frequency domain.

**Contributions:**

- We train a parameter-conditioned FNO for the FHN system, using feature-wise linear modulation to add $\boldsymbol{\lambda} = (D_u, D_v, a, b, \tau)$ into every Fourier layer. A single network covers the full 5D parameter space, whereas prior reaction-diffusion work [6] fixed the physical parameters.
- A bifurcation analysis of the ODE shows that the sampled parameter ranges sit clearly in the oscillatory and excitable regimes of firing neurons.
- We characterize parameter-wise error sensitivity across the 5D space, and demonstrate relatively uniform generalization, with $|\rho| < 0.36$ for all parameter-error correlations.
- We quantify extrapolation behavior outside the training ranges and identify diffusion magnitude as the primary failure mode, while the surrogate extrapolates accurately along the reaction and time-scale axes.
- Within the training parameter range, the surrogate holds sub-$0.1\%$ relative $L^2$ error on both fields while running nearly three orders of magnitude faster than the finite-difference baseline. This positions the surrogate as a fast, differentiable forward model for downstream neuroscience workloads such as inverse parameter inference and closed-loop stimulation, which we motivate and set up but leave to future work.
- We show the same FiLM-conditioned operator extends to the excitable (action-potential) regime, where it reproduces the all-or-none firing threshold and the $c \propto \sqrt{D_u}$ conduction-velocity scaling and replicates full traveling action potentials over 100-step rollouts.

Throughout, we restrict attention to a single spatial dimension. The study is 1D, we leave the extension of the surrogate to the 2D/3D domains to future work.

## 2 Related Work

The FitzHugh-Nagumo model has traditionally been studied with numerical methods such as finite difference methods, finite element methods, and spectral methods [21,12]. These approaches discretize the spatial domain into grids or elements and then iteratively solve the resulting equations. These methods are accurate but require fine spatial and temporal discretization, making large parameter sweeps computationally expensive.

One solution to this problem is to use machine-learned surrogate models instead. Neural operators [11] learn mappings between function spaces, and the Fourier Neural Operator [13] demonstrates this with spectral convolutions. Extensions include physics-informed neural operators [14], multi-resolution methods [15], and equation-free local operators [3].

### 2.1 FNOs for Reaction-Diffusion Systems

The closest prior work, Hao and Song [6], applies FNOs to the Surface Quasi-Geostrophic and Gray-Scott systems, but (i) holds physical parameters fixed, (ii) targets smooth field evolution, and (iii) evaluates generalization only across initial conditions. Our system adds stiff dynamics, sharper activations, and a five-dimensional parameter space that controls distinct ODE bifurcations.

## 3 Mathematical Background

### 3.1 The FitzHugh-Nagumo System

The FitzHugh-Nagumo (FHN) system models neuronal voltage and recovery dynamics through two coupled reaction-diffusion equations [5,18]:

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u + u - \frac{u^3}{3} - v, \tag{1}$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + \frac{1}{\tau}(u + a - bv), \tag{2}$$

where $u(\boldsymbol{x}, t)$ is the membrane voltage, $v(\boldsymbol{x}, t)$ is the recovery variable that gates the slow current, $\boldsymbol{x} \in \Omega \subseteq \mathbb{R}^1$ is the spatial coordinate, and $t \in [0, T]$ is time. The parameter vector $\boldsymbol{\lambda} = (D_u, D_v, a, b, \tau) \in \mathbb{R}^5$ characterizes the system, where $D_u, D_v > 0$ are diffusion coefficients and $(a, b, \tau)$ control the reaction kinetics. Different combinations of $\boldsymbol{\lambda}$ place the system in qualitatively distinct dynamical regimes—principally the *oscillatory* (tonic-firing) and *excitable* (single action-potential) regimes this paper models. Before generating data, we verify via the bifurcation analysis of Section 4.1 that each sampled parameter box lies cleanly inside its intended regime.

### 3.2 Learning Parametric Solution Operators with FNOs

We consider the challenge of learning nonlinear parametric operators $\mathcal{G}_{\boldsymbol{\lambda}} : \mathcal{U} \times \mathbb{R}^5 \to \mathcal{V}$ that map initial states to future states of the FHN system across the parameter space. These operators define mappings between infinite-dimensional function spaces $\mathcal{U}$ and $\mathcal{V}$ for each parameter configuration $\boldsymbol{\lambda} \in \mathbb{R}^5$. The input functions $\boldsymbol{u}_0 = (u_0, v_0) : \Omega \to \mathbb{R}^2$ represent initial conditions, which are transformed by the operator into solution fields $\boldsymbol{u}_{\Delta t} = \mathcal{G}_{\boldsymbol{\lambda}}[\boldsymbol{u}_0] : \Omega \to \mathbb{R}^2$ at time $t = \Delta t$.

**Single-step operator:** Maps the state at time $t$ to the state at time $t + \Delta t$:

$$\boldsymbol{u}_{t+\Delta t}(\boldsymbol{x}, t) = \mathcal{G}_{\boldsymbol{\lambda}}^{\Delta t}[\boldsymbol{u}_t](\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega. \tag{3}$$

For longer time predictions, the single-step operator is applied iteratively:

$$\boldsymbol{u}_{n\Delta t} = \underbrace{\mathcal{G}_{\boldsymbol{\lambda}}^{\Delta t} \circ \mathcal{G}_{\boldsymbol{\lambda}}^{\Delta t} \circ \cdots \circ \mathcal{G}_{\boldsymbol{\lambda}}^{\Delta t}}_{n \text{ times}}[\boldsymbol{u}_0]. \tag{4}$$

### 3.3 Fourier Neural Operator Architecture

The Fourier Neural Operator [13] approximates the solution operator $\mathcal{G}_{\boldsymbol{\lambda}}$ through a neural network $\mathcal{G}_{\boldsymbol{\theta},\boldsymbol{\lambda}}$, where $\boldsymbol{\theta}$ denotes the learnable weights. The architecture consists of three main components:

**1. Lifting:** An initial projection $\mathcal{P} : \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^w$ maps the input channels (here $d_{\text{in}} = 2$ for $(u, v)$) to a higher-dimensional representation of width $w$:

$$\boldsymbol{z}_0(\boldsymbol{x}) = \mathcal{P}(\boldsymbol{u}_0(\boldsymbol{x})). \tag{5}$$

**2. Fourier Layers:** The core of the FNO consists of $L$ Fourier layers. Each layer $\ell \in \{1, \dots, L\}$ applies a convolution and then a nonlinearity:

$$\boldsymbol{z}_\ell(\boldsymbol{x}) = \sigma\left(\mathcal{K}_\ell[\boldsymbol{z}_{\ell-1}](\boldsymbol{x}) + \mathcal{W}_\ell(\boldsymbol{z}_{\ell-1}(\boldsymbol{x}))\right), \tag{6}$$

where $\sigma$ is a nonlinear activation function, $\mathcal{W}_\ell$ is a linear transformation (implemented as $1 \times 1$ convolution), and $\mathcal{K}_\ell$ is the convolution operator.

The convolution $\mathcal{K}_\ell$ operates by transforming to the frequency domain, multiplying with learnable weights, and transforming back:

$$\mathcal{K}_\ell[\boldsymbol{z}](\boldsymbol{x}) = \mathcal{F}^{-1}\left(\boldsymbol{R}_\ell \cdot (\mathcal{F}\boldsymbol{z})\right)(\boldsymbol{x}), \tag{7}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the Fourier transform and its inverse, while $\boldsymbol{R}_\ell \in \mathbb{C}^{w \times w \times k_{\max}}$ are learnable weights [13].

**3. Projection:** A projection $\mathcal{Q} : \mathbb{R}^w \to \mathbb{R}^{d_{\text{out}}}$ maps from the hidden dimension back to the output space ($d_{\text{out}} = 2$ for $(u, v)$):

$$\boldsymbol{u}_{\Delta t}(\boldsymbol{x}) = \mathcal{Q}(\boldsymbol{z}_L(\boldsymbol{x})). \tag{8}$$

### 3.4 Training Objective

Given a dataset $\mathcal{D} = \{(\boldsymbol{u}_0^{(i)}, \boldsymbol{u}_{\Delta t}^{(i)}, \boldsymbol{\lambda}^{(i)})\}_{i=1}^N$ of initial conditions, solutions, and parameters, we minimize the mean squared error:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N} \|\mathcal{G}_{\boldsymbol{\theta},\boldsymbol{\lambda}^{(i)}}[\boldsymbol{u}_0^{(i)}] - \boldsymbol{u}_{\Delta t}^{(i)}\|_{L^2(\Omega)}^2, \tag{9}$$

$\boldsymbol{\theta}$ is optimized using AdamW with learning rate scheduling [16].

# 4 Implementation

## 4.1 Dynamical Regimes and Parameter-Box Placement

To verify that the surrogate is trained on meaningful dynamics, we analyze the spatially-homogeneous reduction of (1)–(2). One fixed point satisfies $u^\star - (u^\star)^3/3 = v^\star$ and $u^\star + a = bv^\star$, which yields the cubic $b(u^\star)^3 - 3(b-1)u^\star + 3a = 0$. Each real root gives one fixed point, and its linear Jacobian

$$J = \begin{pmatrix} 1-(u^\star)^2 & -1 \\ 1/\tau & -b/\tau \end{pmatrix} \tag{10}$$

classifies the local flow. The regime is excitable when there is a single fixed point with $\operatorname{tr} J < 0$ and $\det J > 0$ (a stable rest state), and oscillatory when there is a single fixed point with $\operatorname{tr} J > 0$ and $\det J > 0$, an unstable focus surrounded by a limit cycle. It is bistable when there are three real fixed points.

Figure 1 maps these regimes on the $(a, b)$ plane at the median training $\tau$ and overlays the two parameter boxes this paper models. The first surrogate's box (Section 4.2) clearly sits in the oscillatory region: a finer sweep over $\tau \in [1, 20]$ confirms $100\%$ of the sampled cube is oscillatory, so the surrogate is trained specifically on limit-cycle neurons firing tonically rather than a dormant rest state. These same criteria place the box of the excitable surrogate (Section 5.6, $|u^\star| > 1$) entirely in the stable-rest region. Thus, a single analysis defines both dynamical regimes this paper models.

## 4.2 Data Generation

We generate training and validation datasets by solving the FHN system (1)-(2) with a semi-implicit finite-difference method on a uniform periodic grid. We use second-order central differences for the Laplacian and treat the stiff diffusion term implicitly while the reaction term is explicit. Because the implicit operator $\mathbf{I} - \Delta t\, D\mathbf{L}$ is constant for a fixed parameter set, we precompute its LU factorization once per trajectory and reuse it at every step.

**Initial Conditions** We employ Gaussian Random Fields (GRF) as the standard initial condition type. In 1D, we compute Fourier coefficients according to:

$$\hat{u}_k = \mathcal{N}(0,1) \cdot (1 + |k|^2)^{-\alpha/2}, \tag{11}$$

$$\hat{v}_k = \mathcal{N}(0,1) \cdot (1 + |k|^2)^{-\alpha/2} \cdot 0.5, \tag{12}$$

where $k$ is the wavenumber, $\mathcal{N}(0,1)$ denotes a standard normal random variable, and $\alpha = 2.0$ controls the spectral decay rate. The spatial fields are obtained via inverse Fourier transform, and we normalize each field. We manually add a 0.5 factor to the $v$ field to ensure the inhibitor variable has a smaller initial amplitude than the activator; this is supported by biophysical considerations [7].

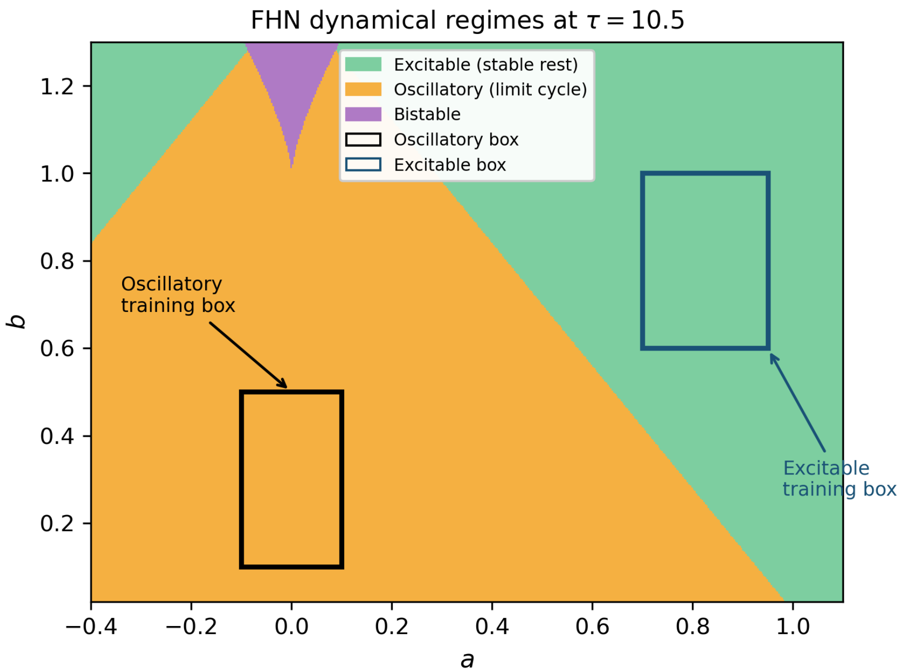



Fig. 1: Dynamical-regime map of the spatially-homogeneous FHN ODE at the median training $\tau = 10.5$. Green: excitable (stable rest state). Orange: oscillatory (limit cycle, tonic firing). Purple: bistable. The black rectangle is the oscillatory training box and the blue rectangle is the excitable training box. It is visually very clear, and supported by a sweep across the full $\tau$ range, that both boxes lie entirely within their respective regimes.

**Parameter Sampling** All five parameters are sampled uniformly from physiologically motivated ranges [5,8,7]: $D_u \sim \mathcal{U}(0.01, 0.1)$ and $D_v \sim \mathcal{U}(0.005, 0.05)$ (voltage diffusing $\sim 2\times$ faster than recovery, which is typical [22,2]), $a \sim \mathcal{U}(-0.1, 0.1)$, $b \sim \mathcal{U}(0.1, 0.5)$, and $\tau \sim \mathcal{U}(1.0, 20.0)$ spans fast recovery to stiff relaxation oscillations. As established in Section 4.1, the entire sampled cube lies in the oscillatory regime of tonically firing neurons. The diffusion bounds span an order of magnitude, from near-pointwise dynamics at the lower bounds to the strong coupling that supports traveling waves at the upper bounds, with the ratio $D_u/D_v \approx 2$ that is common in FHN [22,2]. Because $a$ is near zero, the fixed point is near the cubic nullcline's inflection where the Jacobian trace will produce relaxation oscillations. The recovery strength $b \in [0.1, 0.5]$ stays below the $b \approx 1$ that would restabilize a rest state, and $\tau \in [1, 20]$ spans near-equal to stiff fast/slow time scales [7,23].

**Dataset Specifications** The primary dataset has $N = 8000$ trajectories on a periodic domain $\Omega = [0, 1]$ with $n_x = 256$ points, each evolved to $T = 1.0$ at $\Delta t = 0.01$ with $n_{\text{save}} = 50$ saved snapshots ($\alpha = 2.0$ GRF initial conditions). This yields 50 single-step pairs $(u_t, v_t) \rightarrow (u_{t+\Delta t}, v_{t+\Delta t})$ per trajectory under an 80-20 trajectory-level split (6,400 train / 1,600 validation) used for training and model selection. All final oscillatory-regime metrics in Section 5 are instead reported on a separately generated, identically distributed held-out test set of $1{,}600$ trajectories, disjoint from both the training and validation data and unseen during training and hyperparameter selection. The excitable-regime experiments (Section 5.6) use a separately generated dataset on a longer domain $L = 8$, with $N = 1{,}250$ trajectories ($1{,}000$ train / $250$ validation), $T = 40$, $\Delta t = 0.02$, and $n_{\text{save}} = 100$, and localized super-threshold Gaussian-bump perturbations of the rest state as initial conditions rather than $\alpha = 2.0$ GRF fields.

### 4.3 Model Architecture

We implement the FNO architecture as follows: the network retains the lowest $k_{\max} = 16$ Fourier modes in the convolution, uses a hidden dimension of $w = 64$, and consists of $L = 6$ Fourier layers. The input and output dimensions are both $d_{\text{in}} = d_{\text{out}} = 2$ to accommodate the coupled $(u, v)$ fields. We employ the GELU activation function for its smooth gradient properties, even though the computational cost is higher than the traditional ReLU activation function.

**Lifting and Projection Networks** The lifting network $\mathcal{P}$ elevates the two-channel input to the $w$-dimensional hidden space through the following architecture:

$$\mathcal{P}(\boldsymbol{u}_0) = \text{Conv1D}_{64}\left(\text{GELU}\left(\text{Conv1D}_{128}(\boldsymbol{u}_0)\right)\right), \tag{13}$$

where $\text{Conv1D}_c$ denotes a kernel-size-1 1D convolution with $c$ output channels: the input is expanded to 128 channels, passed through GELU, then projected to the working width of 64.

Similarly, the projection network $\mathcal{Q}$ mirrors this structure to map from the hidden representation back to the physical $(u, v)$ space:

$$\mathcal{Q}(\boldsymbol{z}_L) = \text{Conv1D}_{2}\left(\text{GELU}\left(\text{Conv1D}_{64}(\boldsymbol{z}_L)\right)\right). \tag{14}$$

To preserve input features during the forward pass, we apply a global residual connection to the final output:

$$\boldsymbol{u}_{\text{pred}} = \mathcal{Q}(\boldsymbol{z}_L) + \beta_{\text{global}}\boldsymbol{u}_0, \tag{15}$$

where $\beta_{\text{global}}$ is a learnable scalar initialized to 0.1. It allows the network to learn small perturbations to the input state rather than reconstructing the entire output from scratch.

**Fourier Layer Design** Each of the $L = 6$ Fourier layers computes the real FFT $\tilde{\boldsymbol{z}} = \text{rfft}(\boldsymbol{z}_{\ell-1})$, retains only the first $k_{\max} = 16$ modes (we zero the rest which acts as an implicit low-pass filter). At the same time a pointwise $1 \times 1$ convolution captures local features, and the two pathways are summed and passed through GELU: $\boldsymbol{z}_{\text{pre}} = \text{GELU}(\boldsymbol{z}_{\text{spectral}} + \text{Conv1D}_{64}(\boldsymbol{z}_{\ell-1}))$.

*Choosing* $k_{\max}$=16. The diffusive terms $D_u\nabla^2 u$ and $D_v\nabla^2 v$ damp Fourier mode $k$ at a rate proportional to $k^2$, so even sharp FHN spikes concentrate their energy in the lowest few wavenumbers. An ablation over $k_{\max} \in \{4, 8, 16, 32, 64\}$ on the full 8000-trajectory dataset (Figure 2) shows validation rel-$L^2$ improving $\sim 4\times$ from $k_{\max} = 4$ to $k_{\max} = 16$ and then plateauing, with $k_{\max} = 64$ statistically indistinguishable from $k_{\max} = 16$ while using $3.4\times$ more parameters. We also note that although $k_{\max} = 32$ has a slightly better error score, the $k_{\max} = 16$ model is more stable during autoregressive rollouts, which is the ultimate use case for the surrogate. Thus, we choose $k_{\max} = 16$ as a sweet spot. We ablate only $k_{\max}$, the axis the diffusive spectrum makes most interpretable. The hidden width $w = 64$ and depth $L = 6$ were fixed to standard FNO values [13] that already reach sub-$0.1\%$ error and were not tuned further, so a fuller width/depth sweep remains future work.

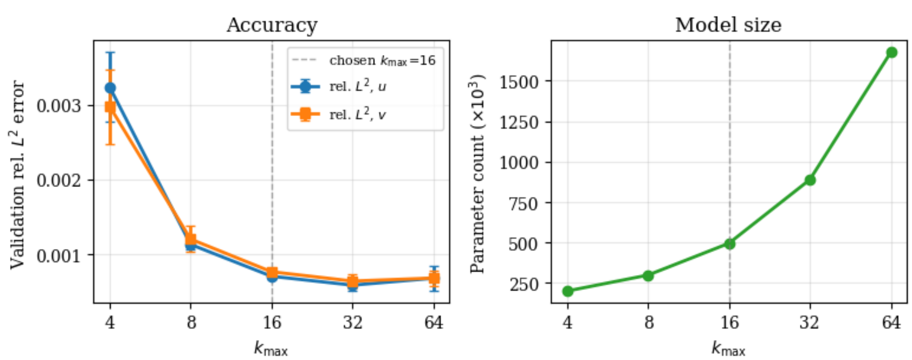



Fig. 2: Ablation of the Fourier-mode truncation $k_{\max}$ on the 8000-sample dataset (3 seeds per configuration). Left: validation rel-$L^2$ on $u$ and $v$. Right: parameter count. $k_{\max} = 16$ lies in the accuracy plateau while keeping model size modest.

**Parameter Conditioning** The lifting, Fourier, and projection blocks described above only take $(u_t, v_t)$ as input, and so they cannot distinguish between different FHN systems. To make it parametric, we condition each Fourier layer on the parameter vector $\boldsymbol{\lambda} = (D_u, D_v, a, b, \tau)$ via feature-wise linear modulation (FiLM) [20].

We first encode $\boldsymbol{\lambda}$ into a width-$w$ feature vector with a three-layer MLP $E$ with LayerNorm (widths $5 \to 2w \to w \to w$):

$$\boldsymbol{h}_{\boldsymbol{\lambda}} = E(\boldsymbol{\lambda}) \in \mathbb{R}^w. \tag{16}$$

At each Fourier layer $\ell$, two independent linear heads produce a per-channel scale $\boldsymbol{\gamma}_\ell \in \mathbb{R}^w$ and shift $\boldsymbol{\beta}_\ell \in \mathbb{R}^w$ from $\boldsymbol{h}_{\boldsymbol{\lambda}}$. The modulation is applied to the Fourier layer's output:

$$\boldsymbol{z}_\ell(\boldsymbol{x}) \leftarrow \boldsymbol{\gamma}_\ell \odot \boldsymbol{z}_\ell(\boldsymbol{x}) + \boldsymbol{\beta}_\ell, \tag{17}$$

where $\odot$ denotes channel-wise multiplication. The encoder $E$ and the per-layer heads are trained with the rest of the network. This turns the single shared operator $\mathcal{G}_{\boldsymbol{\theta}}$ into a $\boldsymbol{\lambda}$-indexed family $\{\mathcal{G}_{\boldsymbol{\theta},\boldsymbol{\lambda}}\}$, with modulation acting the same at every spatial location.

### 4.4 Training and Evaluation

Training pairs $(\boldsymbol{u}_t, \boldsymbol{u}_{t+\Delta t})$ are drawn from trajectories with $u$ and $v$ independently $z$-normalized using statistics from the training set, and each batch also provides the per-trajectory parameter vector $\boldsymbol{\lambda}$. We optimize the per-element MSE over both fields using AdamW [16] with the following parameters: lr $10^{-3}$, weight decay $10^{-4}$, batch 32, gradient clipping 1.0, for up to 1000 epochs on an A100 GPU. We assess accuracy with the relative $L^2$ error per field, MSE, and the autoregressive rollout error obtained by iterating the single-step operator for up to $n_{\text{steps}} = 50$ consecutive steps. The parameter vector $\boldsymbol{\lambda}$ is standardized to zero mean and unit variance before entering the FiLM encoder. Five-seed runs underlie Table 2 and the $\pm$std values in Table 3, and all other tables use seed 42.

| | **Oscillatory** | **Excitable** |
|---|---|---|
| *Data generation* | | |
| Grid $n_x$ | 256 | 256 |
| Domain $L$ | 1.0 | 8.0 |
| Time step $\Delta t$ | 0.01 | 0.02 |
| Horizon $T$ / frames | 1.0 / 50 | 40 / 100 |
| Train / val trajectories | 6400 / 1600 | 1000 / 250 |
| *Architecture* | | |
| Fourier modes $k_{\max}$ | 16 | |
| Hidden width $w$ | 64 | |
| Fourier layers $L$ | 6 | |
| Lift / project | $2 \to 128 \to 64 \,/\, 64 \to 2$ | |
| FiLM encoder | $5 \to 128 \to 64 \to 64$ | |
| Global skip $\beta_{\text{global}}$ init | 0.1 | |
| Parameters | $\sim$0.50 M | |

Table 1: Data-generation, architecture, and training hyperparameters.

## 5 Results

Our evaluation builds a case in the order a downstream user would care about, with each stage answering a distinct question about the surrogate as a forward model. We first establishsingle-step accuracy (is one operator application faithful?), then a per-parameter error breakdown (is that accuracy uniform across the 5D space?), then long-horizon autoregressive rollout (does error stay bounded when the operator is iterated, i.e. the actual use case?), then computational efficiency and finally generalization outside the training ranges and transfer to the excitable regime. Unless stated otherwise, the results in this section concern the oscillatory (tonic-firing) regime; the complementary excitable regime is treated separately in Section 5.6. Model and hyperparameter selection (e.g. the $k_{\max}$ ablation of Section 4.3) used the validation split, whereas all errors reported here are computed on the disjoint held-out test set of $1{,}600$ trajectories described in Section 4.2, whose parameter combinations and initial conditions were unseen during training and selection.

### 5.1 Single-Step Prediction Accuracy

We begin by evaluating the FNO's ability to predict single time steps. Table 2 summarizes the quantitative performance metrics averaged over the entire test set.

| Metric | Activator ($u$) | Inhibitor ($v$) |
|---|---|---|
| Rel. $L^2$ | $6.66 \times 10^{-4}$ | $8.01 \times 10^{-4}$ |
| MSE | $2.98 \times 10^{-6}$ | $9.05 \times 10^{-6}$ |
| MAE | $4.09 \times 10^{-4}$ | $5.86 \times 10^{-4}$ |
| Max AE | $0.178$ | $0.309$ |

Table 2: Single-step prediction accuracy on the held-out test set ($N_{\text{test}} = 1600$ trajectories, $n_x = 256$ spatial points), in normalized space. Values are 5-seed means (per-trajectory mean for relative $L^2$, element-wise averages otherwise). Per-seed variability is reported in Table 3.

| | $\epsilon_{\text{REL}}$ ($\times 10^{-3}$) | | MAE ($\times 10^{-3}$) | | |
|---|---|---|---|---|---|
| Method | $u$ | $v$ | $u$ | $v$ | Params |
| DeepONet | $38.3 \pm 2.0$ | $40.3 \pm 3.4$ | $25.4 \pm 2.1$ | $28.8 \pm 2.5$ | 0.40M |
| Base FNO | $0.859 \pm 0.046$ | $1.038 \pm 0.059$ | $0.798 \pm 0.019$ | $0.898 \pm 0.033$ | 0.63M |
| Param. FNO | $\mathbf{0.666 \pm 0.080}$ | $\mathbf{0.801 \pm 0.140}$ | $\mathbf{0.409 \pm 0.022}$ | $\mathbf{0.586 \pm 0.098}$ | 0.50M |

Table 3: Baseline comparison on the test set ($N_{\text{test}} = 1600$ trajectories, $n_x = 256$). All error values are mean $\pm$ standard deviation across $n_{\text{seeds}} = 5$ independent training runs, reported in units of $10^{-3}$. "Baseline FNO" uses the identical backbone (width 64, 16 modes, 6 layers) but with the parameter vector $\boldsymbol{\lambda}$ broadcast as five constant spatial input channels instead of being injected via FiLM. DeepONet branch uses flattened $(u_t, v_t)$ concatenated with $\boldsymbol{\lambda}$.

The parametric FNO predicts single time steps with sub-$0.1\%$ relative $L^2$ error on both fields: $6.66 \times 10^{-4}$ ($0.067\%$) for the activator $u$ and $8.01 \times 10^{-4}$ ($0.080\%$) for the inhibitor $v$. The inhibitor's marginally higher relative error is expected given its smaller amplitude, so that comparable absolute deviations then translate into larger relative values.

Mean absolute errors ($4.1 \times 10^{-4}$ for $u$, $5.9 \times 10^{-4}$ for $v$) and mean squared errors ($3.0 \times 10^{-6}$, $9.0 \times 10^{-6}$) are physically negligible, and the largest pointwise deviations of max absolute error $0.18$ for $u$ and $0.31$ for $v$ are confined to sharp wavefronts, leaving the total metrics almost entirely unaffected.

**Baseline comparison.** Table 3 compares two natural baselines. DeepONet [17], a branch–trunk operator-learning architecture that encodes the input field and the query coordinates through separate subnetworks, is far worse here: its relative $L^2$ error is roughly $58\times$ higher on $u$ and $50\times$ higher on $v$. The channel-broadcast FNO is more competitive but still performs worse than the FiLM-conditioned model by roughly $30\%$ on both fields, while still containing $26\%$ more parameters ($0.63$M vs $0.50$M). FiLM conditioning therefore improves accuracy and reduces model size relative to broadcasting $\boldsymbol{\lambda}$ as constant channels, so the gain comes from how parameters are added rather than from added capacity.

Qualitative single-step predictions across weakly, moderately, and strongly diffusive regimes are visually indistinguishable from ground truth (Figure 3). The per-regime single-step MAE ranges from $5 \times 10^{-5}$ to $2 \times 10^{-4}$, with the weakly diffusive case having the largest errors, consistent with the diffusion-driven failure mode of Section 5.5. It notes that weaker diffusion leaves sharper wavefronts that are truncated by the $k_{\max}{=}16$ spectral filter.

### 5.2 Parameter Error Correlation

It's important to note that a single averaged error can hide some behavior: a surrogate may look accurate overall while failing along one axis of the parameter space. Because the coordinates of $\boldsymbol{\lambda}$ correspond to distinct ODE bifurcations, the axes where error concentrates carry physical meaning and indicate where to add data augmentation. Thus, we relate the error to each axis of $\boldsymbol{\lambda}$ individually so we can determine whether any single parameter drives accuracy. We note that the Pearson $\rho$ used here captures only linear/monotonic dependence, so a purely nonlinear sensitivity could go undetected. Table 4 reports Pearson correlations between each FHN parameter and the relative $L^2$ error for both $u$ and $v$.

| | $D_u$ | $D_v$ | $a$ | $b$ | $\tau$ |
|---|---|---|---|---|---|
| $u$ ERROR | $+0.143$ | $-0.047$ | $-0.036$ | $+0.033$ | $-0.052$ |
| $v$ ERROR | $-0.166$ | $+0.114$ | $+0.025$ | $-0.027$ | $-0.357$ |

Table 4: Pearson correlation between each FHN parameter and the per-trajectory single-step relative $L^2$ error on the test set. All $|\rho| < 0.36$, indicating relatively uniform generalization across the five-dimensional parameter space.

The largest correlation magnitude anywhere in Table 4 is $|\rho| = 0.357$ (the $\tau$–$v$ term), so every coefficient falls below $0.36$, which indicates relatively uniform accuracy. Contrary to expectation that the faster activator would be more sensitive, $u$ is

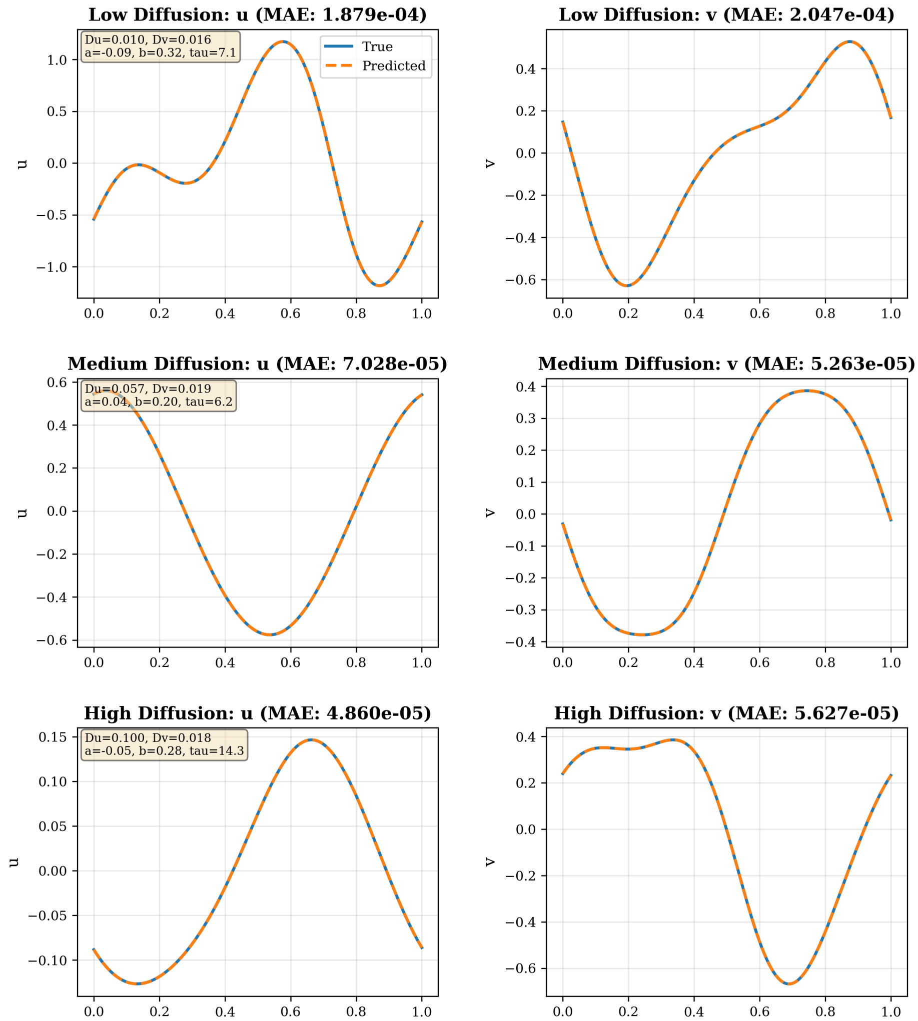



Fig. 3: Single-step prediction across three representative test regimes (rows: weakly, moderately, strongly diffusive). Left: ground truth vs predicted $u$. Center: ground truth vs predicted $v$. Right: absolute error in $u$ and $v$ (scaled for visibility). Predicted and true profiles are visually indistinguishable.

nearly parameter-agnostic: its strongest correlation is a weak $\rho = +0.143$ with $D_u$, and every other coefficient is under $|\rho| \leq 0.06$. The variation that exists is concentrated in the inhibitor $v$, whose dominant axis is the time-scale $\tau$, with $\rho = -0.357$. This is physically sensible, since $\tau$ sets the relaxation rate of the slow recovery variable. The negative sign means $v$ is predicted slightly more accurately at larger $\tau$, where its dynamics are smoother. The reaction parameters $a$ and $b$ are basically uncorrelated with the error on both fields, with $|\rho| \leq 0.05$.

Error thus concentrates along the time-scale axis rather than the reaction axes, but even there the dependence is moderate. This relatively uniform structure indicates that random parameter sampling sufficed to cover the space, and the within-range trend that higher $\tau$ is easier is consistent with the time-scale extrapolation shown next.

### 5.3 Long-Time Autoregressive Rollout

Many applications require predictions over extended time horizons. We evaluate the FNO's autoregressive rollout capability by iteratively applying the single-step operator for $n_{\text{steps}} = 50$ time steps, corresponding to a total evolution time of $T = 0.5$ time units ($50\times$ the training time step $\Delta t = 0.01$).

Most substantially, the operator does not suffer from runaway error accumulation: even after $50$ autoregressive steps the relative $L^2$ error is only $\sim 0.5\%$ for the activator $u$ and $\sim 0.7\%$ for the inhibitor $v$. The activator error grows roughly linearly, whereas the inhibitor error drops over the first few steps before increasing sub-linearly. This early dip reflects the inhibitor's slow time scale, as the single-step operator slightly over-sharpens $v$, but $v$'s own slow, diffusive relaxation damps that transient over the next few steps before the regular error accumulation takes over. Because the activator error grows faster, the two curves converge by step $50$ and $u$ would likely surpass $v$ over a longer time horizon.

Note that over longer time horizons the FNO error could grow to concerning levels. Future work could explore training strategies such as fixed-horizon rollouts to improve long-time stability. A phase-space analysis (Figure 4) confirms that the FNO preserves the limit-cycle geometry at fixed spatial locations, with the only departure being an extremely slow phase drift consistent with the autoregressive error accumulation.

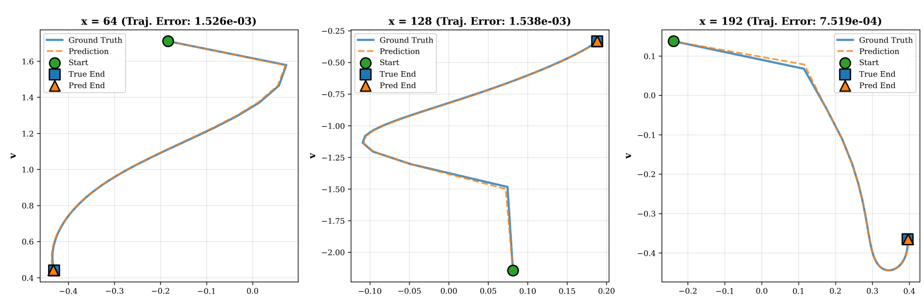



Fig. 4: Phase-space portraits at three spatial locations ($v$ vs $u$, 50 time steps). Blue: ground truth, Orange dashed: FNO. Green circle: initial condition. Square and triangle: ground-truth and predicted endpoints, respectively. The operator traces the same limit-cycle geometry as the solver, with only a gradual end-of-rollout phase drift.

### 5.4 Computational Efficiency

Table 5 compares the FNO against the semi-implicit FD solver on one 50-step trajectory rollout. The FNO delivers $111\times$ speedup at batch 1 and $920\times$ at batch 32, turning a 2-second simulation into 2 ms. There is, however, a tradeoff: $\sim$150 MB for the FNO vs 8 MB for the FD solver's sparse matrices. However, we think this is a reasonable price for the speedup.

| Method | Time/step | Total (50 steps) | Memory |
|---|---|---|---|
| FD (batch 32) | 42.3 ms | 2.12 s | 8.4 MB |
| FNO (batch 1) | 0.38 ms | 0.019 s | 147 MB |
| FNO (batch 32) | 0.045 ms | 0.0023 s | 152 MB |
| Speedup | $111\times$ / $940\times$ | $111\times$ / $920\times$ | — |

Table 5: Computational efficiency comparison between FNO and finite-difference (FD) solver for a single trajectory rollout ($n_x = 256$, $n_{\text{steps}} = 50$, NVIDIA A100).

### 5.5 Generalization to Unseen Regimes

We evaluate the FNO on parameters outside the training ranges to determine its extrapolation capabilities. Table 6 reports four scenarios.

| Parameter Regime | $\epsilon_{\text{REL}}(u)$ | $\epsilon_{\text{REL}}(v)$ |
|---|---|---|
| Within range (test) | 0.0006655 | 0.0008012 |
| Low diffusion ($D_u$=0.005, $D_v$=0.002) | 0.007085 | 0.008089 |
| High diffusion ($D_u$=0.15, $D_v$=0.08) | 0.03401 | 0.01635 |
| High time-scale separation ($\tau$=30) | 0.0005277 | 0.001066 |
| Low recovery ($b$=0.05) | 0.0003559 | 0.0003577 |

Table 6: Generalization performance outside the training ranges. Relative $L^2$ error, mean over 100 test trajectories per scenario.

Diffusion magnitude is the primary failure mode. When both diffusion coefficients are 50–60% above the training range, the activator error increases to $3.4\%$ (roughly $51\times$ the within-range value) and the inhibitor to $1.6\%$ ($\sim 20\times$), while lowering them below the training range gives only a $\sim 10\times$ increase. In contrast, the model extrapolates well along the reaction and time-scale axes: while at $\tau$=30, the activator error stays below the within-range value and the inhibitor rises only by $\sim 33\%$, and the low-recovery case ($b$=0.05) is more accurate than within-range on both fields. This contradicts the idea that the stiff time-scale separation regime is the difficult one. Instead, the sharp spatial gradients, which shift energy into the high wavenumbers that become truncated by the spectral layers, are what degrade accuracy. This suggests that future work could improve extrapolation by augmenting the training data with sharper profiles.

### 5.6 Excitable Regime: Action-Potential Propagation

The experiments above target the oscillatory regime. A complementary and historically defining regime is the excitable one: a single stable rest state from which a sufficiently strong localized stimulus launches a traveling action potential. It exhibits two phenomena absent from the limit-cycle regime, an all-or-none firing threshold and a diffusion-controlled conduction velocity, and we test whether the same FiLM-conditioned operator learns them.

We train a second, architecturally identical surrogate (Table 1) on a parameter box that Section 4.1 places entirely in the excitable region. A fixed point is excitable whenever $|u^\star| > 1$, and sampling $D_u \in [0.006, 0.030]$, $D_v \in [0.0005, 0.005]$, $a \in [0.70, 0.95]$, $b \in [0.60, 1.00]$, $\tau \in [8, 20]$ yields $|u^\star| \geq 1.10$ for all $5\times10^4$ Monte-Carlo evaluations. Pulses are launched from rest by a super-threshold Gaussian bump on a domain $L = 8$, which is long enough to keep the counter-propagating fronts from colliding with each other.

Single-step accuracy matches the oscillatory model (Table 7): relative $L^2 \leq 2 \times 10^{-4}$ on both fields. A 100-step rollout over the full $T = 40$ horizon keeps relative $L^2$ below $0.9\%$, thus the model is able to serve as a surrogate for the entire traveling pulse rather than simply its onset (Figure 6).

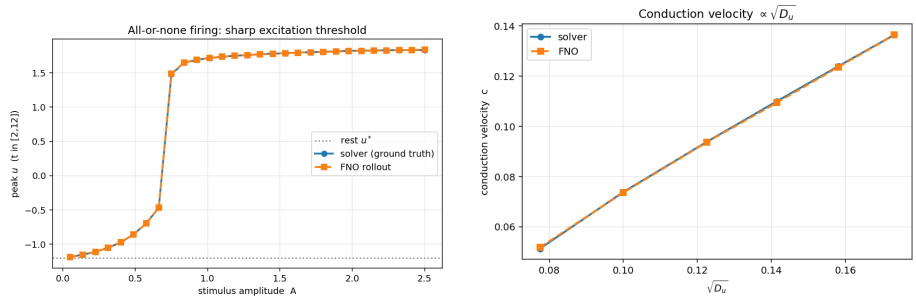
| METRIC | ACTIVATOR ($u$) | INHIBITOR ($v$) |
|---|---|---|
| RELATIVE $L^2$ | $1.97 \times 10^{-4}$ | $1.87 \times 10^{-4}$ |
| MSE | $1.5 \times 10^{-7}$ | $1.7 \times 10^{-8}$ |
| MAE | $1.8 \times 10^{-4}$ | $6.9 \times 10^{-5}$ |
| MAX AE | 0.026 | 0.0039 |

Table 7: Single-step prediction accuracy in the excitable regime, $N = 250$ trajectories, in normalized units.

Both excitable-media signatures match the solver (Figure 5). The all-or-none threshold is visible as a sharp saddle-node transition at $A^\star \approx 0.7$ Figure 5a, and the conduction velocity follows the predicted $c \propto \sqrt{D_u}$ law [8] across the full diffusion sweep (Figure 5b). Over the full 100-step horizon the operator reproduces both fields of the propagating pulse, with error confined to a thin band at the moving wavefront, so it clearly tracks the live pulse rather than only its onset (Figure 6).

### Limitations and Future Work

Several limitations bound the present study and set up its natural extensions. First, all experiments are one-dimensional; we leave 2D/3D geometries for future work. Second,




(a) All-or-none threshold.

(b) Conduction velocity.

Fig. 5: Quantitative excitable-media signatures, solver vs. FNO rollout. **(a)** Peak post-stimulus voltage vs stimulus amplitude $A$: a sharp threshold ($A^\star \approx 0.7$) separates sub-threshold decay to rest ($u^\star \approx -1.2$) from full action-potential firing ($u \approx +1.8$), and both methods are identical on both branches and at the threshold. **(b)** Traveling-pulse speed scales linearly with $\sqrt{D_u}$ as demonstrated by excitable-media theory, and the FNO matches the solver perfectly.

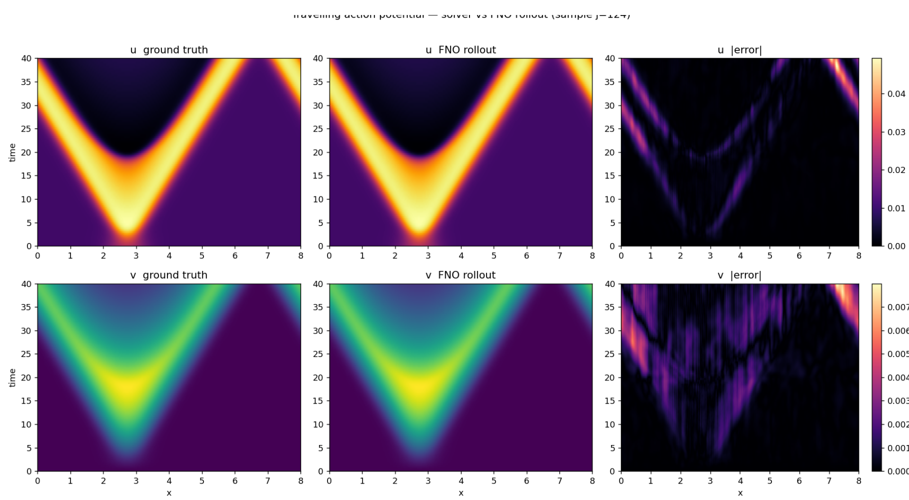



Fig. 6: Traveling action potential, solver vs FNO rollout. Rows: activator $u$ (top), recovery $v$ (bottom). Columns: ground truth, FNO 100-step rollout, absolute error. The propagating wavefront is reproduced over the full $T = 40$ horizon, and error is confined to the moving front.

extrapolation degrades primarily along the diffusion axis (Section 5.5), where sharp wavefronts push energy past the spectral truncation. Third, the parameter-sensitivity screen of Section 5.2 is linear and could miss nonlinear dependencies.

Most importantly, the differentiable surrogate is meant to enable downstream inverse and control tasks that we motivate. A concrete target is signal-based conduction blocking: finding a stimulus $I_{\text{ext}}(\boldsymbol{x}, t)$ that halts action-potential propagation, the basis of kHz nerve block [9]. We establish two prerequisites for this. (i) A differentiable, accurate forward surrogate: because the FNO is differentiable in $\boldsymbol{\lambda}$ and its inputs, such a stimulus becomes a gradient-descent target, with the search direction obtained in a single backward pass through the rollout, whereas the numerical solver requires a separate integration for every candidate. (ii) Parameter robustness: since the FiLM conditioning already spans the parameter family, a candidate signal can be stress-tested for robustness across $\boldsymbol{\lambda}$. Demonstrating an actual recovered-parameter or optimized-stimulus result is the natural next step, moving the surrogate toward equation-free, system-level neuromodulation design [3].

## Ethical Statement

Training neural operators is computationally intensive and carries a notable environmental footprint. While this FNO model is far smaller than state-of-the-art systems, the concern remains: the cumulative energy used in hyperparameter tuning, model training, and extensive simulations can be non-negligible. We plan to mitigate the environmental impact by limiting the number of training runs and using mixed-precision and tuned batch sizes to maximize hardware utilization [19].

In addition, over-reliance on AI-generated approximations in high-stakes domains can be dangerous. If FNO modeling of the FHN system is used as a building block for general research (e.g., building more refined neuronal models), and researchers place blind trust in the predictions, it may lead to incorrect features or research based on false data. Overdependence on AI tools can lower human expertise and become an issue when the AI fails or is used outside its scope, especially in the medical field [10]. Thus, the FNO models should augment decision making rather than replace it.

Finally, the bias in modeling from training data or underlying design is another ethical concern. In the context of FNOs for the FitzHugh-Nagumo model, a form of bias could arise if the training dataset of simulated scenarios is not sufficiently representative of all relevant conditions (for example, if all training simulations use a narrow range of model parameters or initial conditions). It is known that when training data are unrepresentative or incomplete, the learned model will yield biased outputs that systematically err on those underrepresented conditions [4]. If the model is applied in a biomedical context, it should be cross-checked to make sure it does not inadvertently perpetuate any biases that could lead to health disparities, i.e., differing accuracy on data from different patient groups.

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